Pham Viet Hoang

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Experience

Netcompany Group A/S, Junior Software Engineer **Feb 2025 – Feb 2026**

Amplio Platform — Internal platform for building custom tax management, lending, and banking solutions.

- Implemented Elasticsearch-based retrieval of user and "workflow instance" metadata across **500K+ records**, using the Outbox Pattern, Debezium, and Kafka, with performance achieved under **200ms** for query latency.
- Redesigned distributed and multi-level cache using Redis and Caffeine, handling **10K QPS** with **436ms** P95 latency.
- Developed daily batch jobs to migrate nearly **1,000** images from completed or aborted "workflow instances" in database to Amazon S3, **reducing the storage overhead**.
- Automated certificate-expiry monitoring system through daily scheduled-check, **eliminating manual tracking**.
- Optimized **10+ bulk queries** using batch processing to reduce database load and improve query performance.
- Built a cd pipeline to transfer documents from repos to SP Toolkit, reducing view downtime from **20 mins to 1s**.
- Led 2 teams of 5 people in 2-month period, delivering project onboarding, resolving **60** bugs, and releasing **10** features.
- **Tech Stacks**: Spring Boot, React, S3, Elasticsearch, Debezium, Kafka, Redis, Caffein, PostgreSQL, CI/CD.

Payoo VietUnionGroup - HCMUS Graduation, Intern Software Engineer **Jan 2024 – Aug 2024**

Ticket Management System — An application for managing movie ticket bookings, sales, and purchases.

- Collaborated in a 2-member team to design a **microservices-based platform** using **Spring Boot** and **DDD**, developing order service, customer and movie management, and **VNPay** payment integration.
- Implemented distributed transaction workflows using the **Saga Pattern** and **Axon Framework**, incorporating **idempotency** mechanisms to prevent duplicate processing and ensure reliable transaction handling across services.
- Designed PostgreSQL schemas and managed Flyway migrations to ensure database consistency across environments.
- Contributed to building **Grafana dashboards** to visualize application metrics and improve system observability.
- Integrated **Logstash** for centralized log collection and documented RESTful APIs using **Swagger**.
- Built an API Gateway integrated with Keycloak for centralized request routing and authentication management.
- **Tech Stacks**: Spring Boot, Axon Framework, Grafana, Logstash, PostgreSQL, Docker, Swagger, VNPAY, Keycloak.

Projects

Movies Streaming Platform **Overview**

- Developed a movie streaming platform using Spring Boot, React, and HLS with adaptive bitrate streaming, PostgreSQL metadata storage, S3 media storage, and CloudFront CDN, targeting **sub-3-second** playback startup.
- Built a CDC-driven search and auto suggestion pipeline using Debezium, Kafka, and Elasticsearch with CQRS, achieving **sub-2-second** index synchronization and **sub-200ms P95** search latency under **3000** queries/second.

Shorten URL Application **Overview**

- Designed a URL shortener with Go and PostgreSQL, using Redis caching and pub/sub and Nginx for request routing.
- Implemented worker-based URL persistence, handling **1,000**+ bookmarks per import without blocking API requests.
- Maintained **95**%+ test coverage with nearly **15** integration tests; automated pre-merge testing and Docker Hub image publishing via GitHub Actions, followed by VPS deployment, and New Relic for application monitoring.

Saigon Rush **Overview**

- Integrated Antigravity with Addy Osmani's Agent Skills to streamline planning, coding, testing, and code reviews.
- Used Claude Opus 5.5, PRD workflows, AI coding agents to speed up development and debugging for a 3D game.

Education

University of Science, VNU-HCM **Graduated 2026**

Bachelor of Science in Computer Science *Advanced Program in Computer Science*

- GPA: 8.2/10.0
- Language Proficiency: IELTS 7.0 overall (with a 7.5 in Speaking)

Honors & Awards

Second Prize, The 2020 ICPC Asia Can Tho Regional Contest (2020, 2021) (placed 12th/113 teams) **Link**

Consolation Prize, Vietnam Olympiad in Informatics (2019, 2020) **Link**

First Prize, Ho Chi Minh City Informatics Competition for Gifted Students (2019, 2020)

Technical Skills

Languages: Java, SQL, C++, Go, JavaScript, TypeScript, HTML, CSS.

Frameworks and Skills: Spring Boot, Axon Framework, React, Kafka, Nginx, Swagger, HLS, VNPAY, API Gateway, Keycloak.

Databases, Caching and Cloud: PostgreSQL, Redis, Caffeine, Amazon S3, CloudFront, Elasticsearch.

DevOps & Monitoring: Docker, GitHub Actions, CI/CD, Azure DevOps, Prometheus, Tempo, Grafana, Logstash, New Relic.

AI & Development Tools: Google Antigravity, Claude Opus 5.5.